

Agent Architecture

Autonomous multi-agent orchestration architecture that converts raw municipal archives into deterministic, decision-maker-ready acquisition opportunities.



Master Orchestrator: Autonomous DAG Execution Layer

Gieni operates as a directed acyclic graph (DAG) of nine specialized autonomous agents overseen by the Master Orchestrator. The orchestrator enforces strict data contracts, manages asynchronous event streams, dispatches tasks, and routes edge cases to human-in-the-loop review queues when confidence scores breach minimum delivery thresholds.

1. Master Agent Interaction Diagram



2. The Nine Specialized Autonomous Agents



Agent 1: Property Identity Agent (PIA)

- Purpose: Discovers all physical real estate holdings belonging to a decedent estate and performs initial portfolio classification.
- Inputs: Probate petition text, decedent name, last known residence, tax ID aliases.
- Outputs: Candidate parcel inventory, asset type tags (Homestead, Rental, Acreage, Contiguous Lot).
- Dependencies: County court petition scrapers.
- Human Review: Triggered if estate holds 3+ non-contiguous properties or unaddressed parcels.
- KPIs: Parcel identification coverage (target >85%), false negative rate (<3%).
- Failure Modes: Common name collisions; fallback to probate petition asset schedules and voter registry addresses.



Agent 2: Parcel Reconciliation Agent (PRA)

- Purpose: Establishes high-confidence attribution between decedent identity and specific county tax assessor GIS parcel records.
- Inputs: Candidate parcels from PIA, county GIS layers, tax assessor rolls.
- Outputs: Matched APN/PIN, legal description, situs address, Parcel Attribution Score (PAS, 0–100).
- Dependencies: Property Identity Agent.
- Human Review: Required if PAS falls below delivery threshold (<70 pts).
- KPIs: Attribution accuracy (target >95%), unbundled parcel detection rate (>98%).
- Failure Modes: Out-of-county property ownership; fallback to statewide grantor/grantee index.



Agent 3: Ownership Intelligence Agent (OIA)

- Purpose: Audits deed history to identify legal title vesting, encumbrance liabilities, and Net Equity.
- Inputs: Reconciled APN, county recorder conveyance deeds, open mortgage notes, tax lien records.
- Outputs: Vesting classification (Sole Fee Simple, JTWROS, Trust, TIC, Heir Property), Ownership Complexity Score (10–100), Net Equity Waterfall.
- Dependencies: Parcel Reconciliation Agent.
- Human Review: Mandatory for multi-generational unprobated deeds or unrecorded trust instruments.
- KPIs: Deed instrument retrieval rate (>98%), net equity estimation precision ($\pm 10\%$).
- Failure Modes: Unrecorded deeds or wild deeds; fallback to title company plant index.



Agent 4: Control Intelligence Agent (CIA)

- Purpose: Decouples ownership from decision-making; isolates the de-facto human controller driving property disposition.
- Inputs: Heirs schedules, probate petition petitioner data, physical occupancy indicators, utility account metadata.
- Outputs: Control Archetype (Unified, Informal Leader, Proxy, Trust Fiduciary, Contested), primary decision-maker dossier.
- Dependencies: Ownership Intelligence Agent.
- Human Review: Triggered by hostile family caveats or competing sibling petitions.
- KPIs: Correct decision-maker attribution rate (>85%), outreach adoption rate (>80%).
- Failure Modes: Estranged multi-heir silence; fallback to geographical proximity heuristic.



Agent 5: Authority Resolution Agent (ARA)

- Purpose: Determines the legal authority pathway and statutory signatory capacity under probate court dockets.
- Inputs: Court minute orders, Letters Testamentary/Administration, will codicils, estate attorney records.
- Outputs: Four-Tier Authority Classification (Tiers 1–4), court oversight model (Independent vs. Dependent), gatekeeper-bypassed contact profiles.
- Dependencies: Control Intelligence Agent & Court Docket Feeds.
- Human Review: Required for dependent administration sales requiring court confirmation.
- KPIs: Authority accuracy rate (>90%), attorney quarantine success rate (>98%).
- Failure Modes: Court docket scanning lag; fallback to automated clerk calendar scraper.



Agent 6: Opportunity Scoring Agent (OSA)

- Purpose: Synthesizes financial upside against the Deal Friction Score (DFS) to assign an objective 0–100 viability ranking.
- Inputs: Net Equity tier, Authority certainty score, distress multipliers, Ownership Complexity score.
- Outputs: Composite Score (0–100), Priority Band (Priority A, B, C, Disqualified), actionable SLA assignment.
- Dependencies: Ownership Intelligence Agent, Authority Resolution Agent.
- Human Review: Edge cases scoring within ± 2 points of the Priority A threshold (78–82 pts).
- KPIs: Priority A deal conversion rate (>30%), disqualification accuracy (>99%).
- Failure Modes: Incomplete debt data; fallback to conservative maximum encumbrance default.



Agent 7: Quality Control Agent (QCA)

- Purpose: Acts as the pre-delivery gatekeeper, executing multi-point deterministic verification before file certification.
- Inputs: Assembled opportunity records from Agents 1–6.
- Outputs: QC Certification Pass/Fail, audit flag notes, routing to Tasks & Exceptions database.
- Dependencies: Agents 1–6.
- Human Review: Any file failing automated integrity checks (e.g. conflicting deed dates, mismatched parcel address).
- KPIs: Delivery pass rate (>95%), customer-reported defect rate (<2%).
- Failure Modes: Source data conflict; fallback to secondary title plant search.



Agent 8: Reporting Agent (RA)

- Purpose: Compiles certified files into standardized delivery packages and syncs to partner CRM platforms.
- Inputs: QC-passed opportunity records.
- Outputs: Weekly Batch Packages, Urgent Priority A Alerts, partner CRM record creation (Podio, Salesforce, HubSpot).
- Dependencies: Quality Control Agent.
- Human Review: CRM field mapping errors or delivery failure alerts.
- KPIs: Delivery SLA adherence (>99%), webhook dispatch latency (<30s).
- Failure Modes: Partner CRM API rate limits; fallback to encrypted portal drop and SMS alert.



Agent 9: Feedback Agent (FA)

- Purpose: Closes the intelligence loop by ingesting partner disposition telemetry and recalibrating scoring models.
- Inputs: Partner disposition tracking (contacts, appointments, offers, executed contracts, rejections).
- Outputs: Score weight recalibration factors, county-level friction adjustment coefficients, dynamic re-scoring triggers.
- Dependencies: Client Delivery & Disposition Tracking.
- Human Review: Periodic review with Gieni operations team during monthly strategy sessions.
- KPIs: Weekly disposition compliance (>80%), model predictive accuracy gain (>5% per quarter).
- Failure Modes: Missing partner feedback; automated reminders and partner check-in scheduling.

3. Architectural Guardrails: Deterministic Human-in-the-Loop

Gieni enforces a strict boundary between automated agent heuristics and human judgment. Autonomous agents perform 90%+ of data ingestion, reconciliation, and calculations, but exceptions are deterministically

isolated:



Autonomous agents propose; deterministic rules govern; human experts review exceptions; verified outcomes calibrate the next generation of models.